Shiv Narayan Yadav

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Bhopal, Madhya Pradesh

Summary

- GIS & Remote Sensing Specialist with 6+ years of hands-on experience delivering end-to-end geospatial solutions across Conservation, Climate Change, Climate finance, Carbon Credit, Wildlife Management, and Environmental Research Projects. Expert in LULC mapping, Change detection, Lidar data processing and analysis, UAV data processing, multi team geospatial operations, develop custom tools, and produce audit-ready datasets, maps, and dashboards. Adept at transforming complex field and satellite data into actionable insights for decision making, planning, and monitoring.

Employment Highlights

- Currently working as GIS Consultant at Indian Institute of Forest Management, Bhopal in Project TEEB AgriFood Initiative in Assam since August 2026.
- Worked at EarthTree Data Services Pvt. Ltd. as GIS Analyst in Regeneration Meghalaya ARR Project from June 2024 to August 2026.
- Worked at Wildlife Institute of India as a Project Associate - II (GIS) in "Present status of livestock grazing and suggestive plan to phase it out in Tirthan & Sainj Wildlife Sanctuaries" project from July 2023 to December 2023.
- Worked at Wildlife Institute of India as a Project Associate - I (GIS) in "Uttarakhand State Wildlife Population Estimation for the Himalayan Zone" project from April 2021 to June 2023.
- Worked at Wildlife Institute of India as Junior Project Biologist (GIS) in WII-UNDP SECURE Himalaya Project from January 2019 to January 2021.
- Worked at MapMyIndia as a GIS Trainee in Map database product development department From February 2017 to September 2017.
- Worked at Centre for Social Forestry and Eco-Rehabilitation, Allahabad, a unit of Indian Council of Forest Research and Education (Ministry of Environment and Forest) from April 2015 to July 2015 as a Trainee.

Education

- **ITC, University of Twente** — Enschede, The Netherlands
  Postgraduate Diploma in Geo-informatics — *August 2018*
- **Chhatrapati Sahu Ji Maharaj University** — Kanpur, Uttar Pradesh, India
  Master of Science, Environment Science — *July 2016*
- **Chhatrapati Sahu Ji Maharaj University** — Kanpur, Uttar Pradesh, India
  Bachelor of Science, Zoology, Botany & Chemistry — *July 2014*

Skills Summary

- **GIS Data Creation:** Digitization, Geo-referencing, Attribution, Geo-statistical analysis, Querying, Classification, Topological cleaning, Road buffering, Projection, Format conversion, Spatial database handling, Database designing, Process automation, Model Builder etc.
- **Data Processing & Migration:** GIS Data processing, Schema creation, Domains, Subtype and Data Migration.
- **RS Data Processing:** Lidar and UAV data processing and analysis, Image Classification (Supervised & Unsupervised), Image processing, DEM generation with contour, TIN Generation, Geo-referencing, Sub-setting, Mosaic, Feature Extraction, Resolution merge, Pan-sharpen, Contrast stretching etc.
- **Software:** ArcGIS Pro, Agisoft, WebODM, DJI Terra, Erdas Imagine, QGIS, Google Earth Engine, Global Mapper, PostgreSQL, PostGIS, GeoServer.
- **Programming:** Python, JavaScript
- **Soft Skills:** Documentation, presentation, interpersonal and communication, time management

Experience

- **Indian Institute of Forest Management** — Bhopal, Madhya Pradesh
  Consultant — *Aug 2026 - Present*
  - **Project**: The Economics of Ecosystems and Biodiversity (TEEB) AgriFood Initiative in Assam.
  - **Duties and responsibilities**: Processing and analysis of UAV imagery and LiDAR point-cloud data to generate high-resolution orthomosaics, DEMs/DSMs, canopy height models, and vegetation indices, and integrate them with satellite remote sensing and GIS data for precise terrain, biomass, and land-cover assessments.
  - Develop, calibrate, and validate biophysical and environmental models (**DSSAT, SWAT, SWAT-CUP, InVEST**) to assess ecosystem services, agricultural productivity, carbon dynamics, nutrient flows, water resources,

and climate change impacts under alternative land-use and policy scenarios.

- Build spatial databases and thematic maps, and conducted land-use/land-cover analysis, watershed characterization, ecosystem service mapping, and climate vulnerability assessments by integrating climate, soil, hydrological, and socio-economic datasets.
- Analyze large datasets in **Python and R** for model calibration, sensitivity, and uncertainty analysis, and translate the outputs into policy-relevant insights on GHG emissions, carbon sequestration, water availability, and soil health.
- Author technical reports, modelling protocols, policy briefs, and scientific publications. Also collaborate with government agencies and research partners, delivered training workshops, and ensured timely achievement of project milestones.

- **EarthTree Data Services Pvt. Ltd.** Shillong, Meghalaya, India
  GIS Analyst *Jun 2024 - Aug 2026*
  - **Project**: Regeneration Meghalaya ARR Project for Carbon Credit generation
  - **Duties and responsibilities**: Led a team of GIS analysts, ensuring high quality and timely MRV deliverables.
  - Managed ARR geodatabases; designed strata/grids/plots, applied topology QA/QC, and published editable map services using QGIS & GeoServer.
  - Performed Eligibility Area Analysis to identify ARR sites across Meghalaya by integrating RS datasets, legal exclusions, land use history, and spatial criteria using Google Earth Engine (GEE) for multi-temporal analysis.
  - Developed Python, GEE scripts for automating geospatial tasks: MRV tool compatible Attributes generation for Land Registration Boundaries and Farms Designs etc.
  - Produced MRV ready outputs; sampling frames, accuracy metrics, audit compliant tables, and verification grade maps / dashboards and data packs (GeoPackage/CSV/web layers); coordinated with stakeholders/auditors; maintained data security, backups.
  - Coordinated with Drone, Forestry, Community Engagement, Carbon Accounting, and Audit teams to deliver integrated MRV workflows.
  - Software: QGIS, Python, GEE, JavaScript, PostgreSQL, PostGIS, GeoServer, AWS

- **Wildlife Institute of India** Dehradun Uttarakhand, India
  Project Associate - II *July 2023 – December 2023*
  - **Project**: Present status of livestock grazing and suggestive plan to phase it out in Tirthan & Sainj Wildlife Sanctuaries
  - Duties and responsibilities: Undertaken geo-spatial analysis such as Survey data processing, GIS data preparation & processing, LULC classification, Species distribution modelling etc.
  - Developed Python Scripts, queries and developed tools using model builder to support GIS workflow.
  - GIS and RS data collection through various sources such as USGS, Sentinel-hub, NASA LP DAAC, Bhuvan, OSM etc. Undertaken study and survey design in ArcGIS environment.
  - Provided technical coordination and support of geospatial activities between different stakeholders.
  - Undertaken Designing, creating and plotting maps to assist in effective communication and writing of reports. To assist and coordinate stakeholder meetings, preparing presentations and reports related to geospatial and related technological activities.
  - **Software:** ArcGIS, QGIS, Erdas Imagine, Google Earth Engine, Google Earth Pro, Global Mapper, Python, R.

- **Wildlife Institute of India** Dehradun, Uttarakhand, India
  Project Associate - I *April 2021 - June 2023*
  - **Project**: Uttarakhand State Wildlife Population Estimation for the Himalayan Zone.
  - **Duties and responsibilities**: GIS data collection from USGS, sentinel hub, Bhuvan, OSM etc. Undertaken study and survey design in ArcGIS environment.
  - Developed Python Scripts, queries and developed tools using model builder to support GIS workflow.
  - Undertaken Survey data processing, GIS data preparation & processing (Raster & Vector).
  - Performed GIS Data analysis such as LULC classification, Species distribution modelling.
  - Undertaken Designing, creating and plotting maps.
  - Organized workshops for technical training support to the government departments & Report Writing.
  - **Software:** ArcGIS, QGIS, Erdas Imagine, Google Earth Engine, Google Earth Pro, Global Mapper, Python, R.

- **Wildlife Institute of India** Dehradun, Uttarakhand, India
  Junior Project Biologist *Jan 2019 - Jan 2021*
  - **Project:** Identifying, delineating and mapping areas with High Conservation Values (HCV) and developing management recommendations/plans for the SECURE Himalaya landscape in Himachal Pradesh by UNDP.
  - **Duties and responsibilities**: Developed Python Scripts, queries and developed tools using model builder to support GIS workflow.
  - Undertaken Survey data processing and converted in usable GIS format.
  - GIS Data analysis and performed HCV area delineation in ArcGIS environment.

- Undertaken other supporting GIS analyses, e.g. LULC classification, Species distribution modelling etc.
- Provided technical coordination and support of geospatial activities between different government stakeholders, NGOs and local communities in Himachal Pradesh.
- Designing, creating and plotting maps to assist in effective communication and writing of reports.
- To assist and coordinate stakeholder meetings, preparing presentations and reports for geospatial and related technological activities.
- **Software:** ArcGIS, QGIS, Erdas Imagine, Google Earth Engine, Google Earth Pro, Global Mapper, Python, R.

- **MapMyIndia** New Delhi, India
  GIS Trainee *Feb 2017 – Sept 2017*
  - **Project**: Map database product development
  - **Duties and responsibilities**: Geocoding and digitization of map data (Point, Line & Polygons). Existing database update, Data formatting and cleaning (Querying and attribute update).
  - Data Verification and validation by using internet & other sources.
  - **Software:** ArcGIS and MapInfo 15.0

- **Indian Institute of Remote Sensing** Dehradun, Uttarakhand, India
  Dissertation Project *June 2018 - August 2018*
  - **Project**: Assessment of Climate Change Impact on Hydrological Regime of a River Basin in a Geospatial Environment
  - **Workflow:** Data collection, setup suitable hydrological model in a geospatial environment, Hydrological simulation of a basin to study the impact of climate change on its hydrology
  - Development of a Web-based portal to visualize the simulated water balance components of the future for the basin using HTML, JavaScript & CSS.
  - Designed and developed measurement tool, Layers visibility tool and Zoom in and Zoom out tool within the portal using leaflet library.
  - Designed and developed basin information functions and climate model functions using JavaScript.
  - Designed and developed basin button and climatic button using JavaScript
  - **Software**: HTML, JavaScript, CSS, QGIS, SWAT model, leaflet

Academic Projects

- **Galaxy Computer Education and Training Institute** Abu Dhabi, UAE
  Module Project *January 2024*
  - **Project**: Game of sticks using Python.
  - **Workflow:** Developed the game of sticks using python
  - Created a readme and license file.
  - Built a python package index.
- **Computer Education & Technology Promotion Association (CETPA)** Noida, India
  Module Project *December 2018*
  - **Project**: Task Manager for Corporates
  - **Workflow:** Created web pages using HTML and CSS
  - Implemented UI requirements using ASP .NET
  - Created database using MySQL.
- **Centre for Social Forestry and Eco-Rehabilitation** Allahabad, Uttar Pradesh, India
  External Affiliate- Dissertation Research Trainee *April 2015 - July 2015*
  - **Master thesis**: Flora study of Vindhya region Natural Forest of district Mirzapur at Devrikala.
  - **Workflow**: Quadrat data collection, Data analysis and mapping.
  - **Software**: QGIS, Google Earth
- **Indian Institute of Remote Sensing** Dehradun, Uttarakhand, India
  Module Project *April 2018*
  - **Project**: 3D Network Analysis of a Hilly Area (Mussoorie, Dehradun, Uttarakhand) by using GIS
  - **Workflow:** Design and development of 3D Network Database in GIS Environment, Identifying facilities for easy access to the users in emergency cases. Prediction of new optimal routes in an unplanned area.

Training Certificates / Membership

- **Basics of Python programming** Galaxy Computer Education and Training Institute
  Classroom Training Abu Dhabi, *2024*
- **ASP .NET** CETPA

Classroom Training Noida, *2018*

- **International Society for Photogrammetry and Remote Sensing student consortium** ISPRS
  Individual Member *2026*
- **Programming for Geospatial Hydrological Applications, IHE Delft** UNESCO
  Online Course *April*
  2021
- **Google Earth Engine Training** IIRS - ISRO
  Classroom Training *May*
  2018
- **Role of remote sensing & GIS in natural resources management** RSAC, Lucknow
  Classroom Training *March*
  2015

Publications / Reports

- Khali, H., Mishra, A. P., Yadav, S. N., Saber, K., & Pande, C. B. (2024). Temporal Investigation of Chlorophyll-A in the Bhimgoda Barrage and Wetland Landscape Using Remote Sensing and GIS. In Natural Resource Monitoring, Planning and Management Based on Advanced Programming (pp. 115-140). Singapore: Springer Nature Singapore.
- Mishra, A. P., Kumar, A., & Yadav, S. N. (2023). Ecology and conservation of threatened medicinal plants in the Trans-Himalayan region of Nanda Devi Biosphere Reserve, Western Himalaya. Trees, Forests and People, 100451.
- Singh, N., Yadav, S. N., & Lyngdoh, S. (2023). Identifying Conservation Values: a Case Study in Trans-Himalayan Region of Thanpattan, Lahaul-Spiti, Himachal Pradesh. Parks, 29(2), 31–40. https://doi.org/10.2305/IUCN.CH.2023.PARKS-29-2NS.en
- Lyngdoh, S., Sathyakumar, S., Bhatnagar, Y.V., Singh, N. and Yadav, S.N. (2021). Identifying, delineating and mapping areas with high conservation values and developing management recommendations/plans for SECURE Himalaya landscapes in Himachal Pradesh. Technical report. Wildlife Institute of India, Dehradun. 225 pp.
- UKFD - WII (2022). Uttarakhand state macaque and langur population estimation, 2021. Survey Report. Uttarakhand Forest Department and Wildlife Institute of India, Dehradun, 82pp. (Unpublished)
- UKFD - WII (2023). Snow leopard population assessment of India- Uttarakhand State, 2023. Survey Report. Uttarakhand Forest Department and Wildlife Institute of India, Dehradun, 61pp. (Unpublished)
- UKFD - WII (2022). Uttarakhand State Wildlife Population Estimation, Mid-Himalayan Zone (>1,000m & <3,500m), Interim Report, December 2022. Uttarakhand Forest Department and Wildlife Institute of India, Dehradun, 26pp. (Unpublished)

Personal Details

- **Languages**: Hindi, English

References

Mr. Ankit Rawat (Senior Manager)
EarthTree Data Services Pvt. Ltd., Shillong, Meghalaya

Dr S. Sathyakumar (Scientist-G)
Wildlife Institute of India, Dehradun, Uttarakhand